Synthetic Video Generation for Robot Learning Using World Foundation Models

Shrey Jha, Skyler Saucedo, and Frank Tanner

Innodata Inc.
xxx@innodata.com

Abstract. A robot policy trained on a few hours of egocentric household video sees one counter, one bucket, one lighting condition, and cannot be expected to generalize beyond them. We describe a multi-model pipeline built on NVIDIA Cosmos that turns a small real video dataset into a large set of synthetic training clips by varying appearance while preserving the recorded actions. Cosmos Reason produces a structured ontology of each scene: the task, the objects and their roles, and physically grounded variations for each object. A language model turns the ontology into generation prompts and filters implausible variations, yielding about 70 candidates per 15-second clip. Edge, depth, color and segmentation maps, together with a subtraction mask that isolates the surface to be changed, guide Cosmos Transfer to render the variant. Appearance changes work for simpler scenes; complex, multi-texture scenes and action-level changes remain unsolved. From 116 clips the pipeline could produce about 8,120 variants at negligible language-model cost. Whether synthetic appearance data alone makes a robot more robust, or whether action generation must be solved first, is the question the work leaves open.

Keywords: Synthetic data · World foundation models · NVIDIA Cosmos · Egocentric video · Robot learning.

1 Introduction

[Slides 1–2]

Opening framing, from the title and "Main Points" slides.

- Problem: training a robust robotics model from only two to three hours of real egocentric data. Every object, surface and lighting case cannot feasibly be recorded.
- Proposal: use the NVIDIA Cosmos family of world foundation models to scale a limited dataset by generating appearance variants of recorded clips.
- Three claims the paper will support:
  1. A multi-model pipeline on NVIDIA Cosmos can turn a small real dataset into thousands of synthetic training clips.
  2. Changing appearance works for simpler scenes. Complex scenes and action-level changes remain unsolved.

3. Structured reasoning about a scene produces better variations than random augmentation.

– The open question that frames the paper: is synthetic appearance data enough to make a robot more robust, or does action generation have to be solved first?

Contributions, as the deck states them. Pipeline (Sec. 4); ontology-driven variation and prompt principles (Sec. 5); multi-map guidance and mask construction (Sec. 6); qualitative results and their boundary (Sec. 7); a path to closing the gap (Sec. 8).

2 What We Started With and What We Created

[Slides 3–4]



(a) What we started with (b) What we created

Fig. 1: The running example. (a) A person wiping down a table with a bucket and a rag, one of the real clips. (b) The variant produced by the pipeline when asked to change only the countertop to blue. The wall color has changed completely, and blue has bled into the rag.

– The source clip (Fig. 1a): a person wiping a table with a bucket and a rag. Is this one scene enough to train a robust policy? A robot trained only on it might fail the moment the surface color changes or the bucket is swapped.
– Goal: scale this one clip through permutations of the objects and actions observed.
– The variant (Fig. 1b): produced from the same clip by the pipeline, targeting only the countertop color. The model struggles in places: the wall color changed completely and the blue bled into the rag. Sections 6 and 7 say where and why.
– Key point: from one version of the scene to a plausible variation without recording new video.

3 Background: NVIDIA Cosmos

[Slides 5]

- World foundation models (WFMs) are trained on large volumes of real video to model physics, geometry and causality, and to generate physically plausible future states; NVIDIA positions them as simulators that let robots and autonomous vehicles test outcomes before acting [6].
- **Cosmos Predict** [1]: generates future world states as video from text, image or video input (Text2World, Image2World, Video2World); 2B and 14B parameter scales.
- **Cosmos Transfer** [5,2]: adapts video across environments, lighting and domains (Sim2Real, Real2Real) with multi-control conditioning on depth, segmentation and edge maps. Primarily used here for aesthetic transformations.
- **Cosmos Reason** [7]: vision-language model for physical scene understanding; reasons over objects, intent and interactions with chain-of-thought; annotates and critiques synthetic data.
- Cosmos 3 (2026) unifies Predict, Transfer and Reason in a single omni-model built on a Mixture-of-Transformers architecture [3].
- This work uses Transfer and Reason.

4 How We Used World Foundation Models

[Slides 6]

- Original clip prepared with Cosmos Curate.
- The clip is fed to Cosmos Reason, which produces an ontology, a structured understanding of the observed scene (Sec. 5.2).
- The ontology goes to GPT-4o mini, which writes the text prompt in the form Cosmos Transfer expects (Sec. 5.3).
- Four guidance maps are generated to preserve edges, depth and color, plus a segmentation; a subtraction mask restricts where the model may paint (Sec. 6).
- Cosmos Transfer renders the synthetic training video: same task, same actions, new appearance.
- Every input comes from an earlier stage of the pipeline. Nothing is manual except choosing which variation to generate.

5 Part 1: Reasoning with Structured Ontologies

5.1 Ontology-driven variation

[Slides 7]

- Premise: the model must associate objects with the actions and physical context they take part in, not just with how they look.

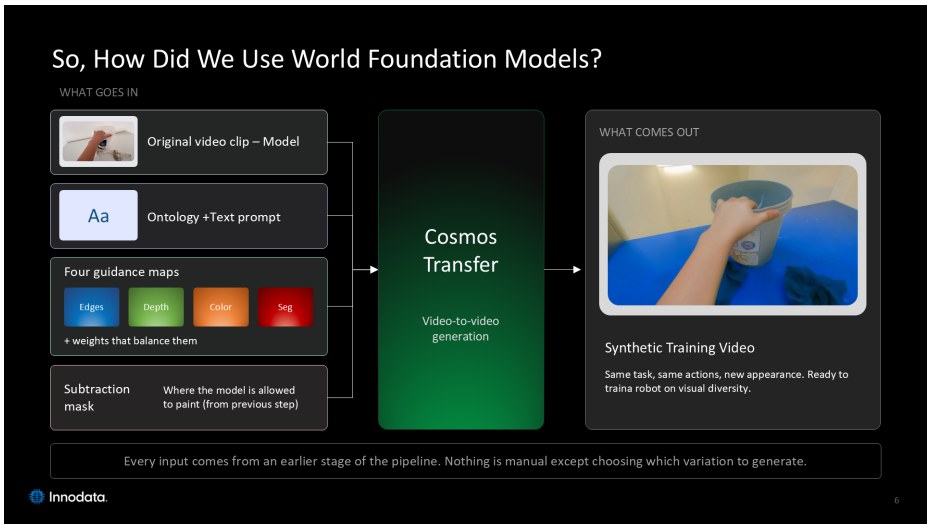


Fig. 2: Pipeline as presented. Inputs on the left: the original clip, the ontology and text prompt, four guidance maps with balancing weights, and a subtraction mask marking where the model may paint. Cosmos Transfer performs video-to-video generation. Output on the right: a synthetic training video with the same task and actions and a new appearance. (Redraw as a vector figure for the final version.)

– Contrast with domain randomization [8], the standard technique in robotics: colors, textures and lighting are swapped at random in the hope that the policy learns to generalize, with no understanding of the scene.
– Ontology-driven variation instead:
  • understands the scene first: objects, states, materials, actions, lighting;
  • links each object to its role in the task, so that a bucket in a cleaning task is a cleaning bucket, which behaves differently from a bucket on a shelf;
  • proposes only physically grounded variations per object;
  • is structured, auditable and filterable.
– Result: about 70 plausible variation candidates per 15-second clip, filtered from more than 100 raw proposals.

5.2 Ontology structure: what Cosmos Reason produces

[Slides 8]

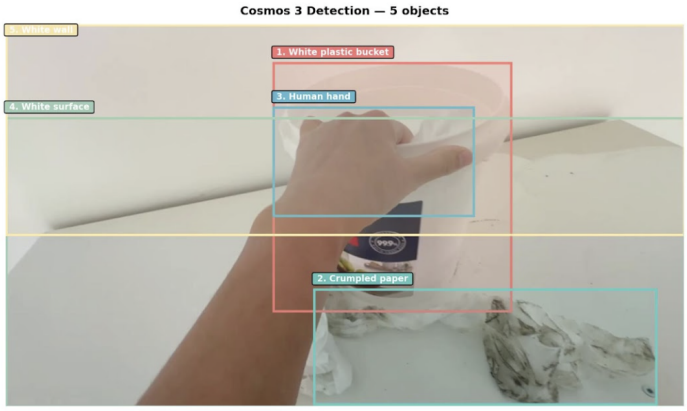


Fig. 3: Detections for the running example: five task-relevant objects with bounding boxes (white wall, white plastic bucket, crumpled paper, human hand, white surface).

– Beyond detection: the ontology records state, material and color for each object and proposes variations grounded in the physical context of the task (Fig. 3).
– Excerpt of the ontology for the running example:

```
"task": {
  "name": "Disposing of crumpled paper into a bucket",
  "phase": "Preparation",
  "action_primitive": "Holding a bucket and positioning it"
},
"key_objects": [{
```

```
  "name": "White plastic bucket",
  "role_in_task": "Container for disposing of waste",
  "robustness_gaps": {
    "what_robot_must_handle": ["Different bucket sizes and shapes",
      "Labels with varying colors and designs", "Wet or slippery buckets"],
    "common_failure_modes": ["Bucket slips from hand", "Bucket tips over",
      "Bucket cannot be opened"]
  }
}]
```

– The model identifies robustness gaps and failure modes such as the bucket tipping over. These are action-level variations, better suited to Cosmos Predict.
– It also proposes appearance variations, such as buckets of different colors, designs, sizes and shapes, which suit Cosmos Transfer.

5.3 From ontology to generation-ready prompts

[Slides 9]

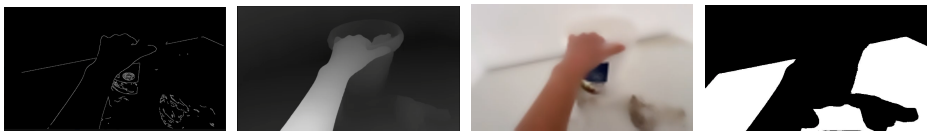
– Input to GPT-4o mini, for the running example: target "white surface being cleaned"; variation "change color to solid blue matte"; preserve bucket, cloth, hands, lighting; context: cleaning task, laminate surface, bright overhead lighting.
– Output, the Cosmos Transfer prompt: *"The video captures a person cleaning a solid blue countertop in a well-lit indoor kitchen. The countertop surface is a rich, vivid blue color with a smooth matte finish. The bucket and paper towels retain their original appearance. . . "*
– Three prompt-engineering principles discovered by experiment:
  1. **Repeat target attributes.** "solid blue countertop" . . . "vivid blue color" . . . "blue countertop surface": threefold reinforcement.
  2. **Explicitly preserve unchanged elements.** "Bucket and paper towels retain their original appearance" protects non-target objects.
  3. **Describe the result, not the change.** "A person cleaning a solid blue countertop," not "change white to blue."
– GPT-4o mini also acts as a filter, rejecting physically implausible variations before they reach Cosmos Transfer.

6 Part 2: How We Guide the Model

6.1 Four maps, four kinds of protection

[Slides 10]

– The model receives four maps with different weights, each protecting a different part of the original video (Fig. 4).
– **Edges**, "keep the shapes": protects the outline of every object in the frame, so the bucket stays a bucket.



(a) Edges (b) Depth (c) Color (d) Mask

Fig. 4: The four guidance maps for one frame of the running example. Edges keep the shapes; depth keeps the layout; color keeps the lighting; the mask says where to paint.

- **Depth**, "keep the layout": protects what is in front and what is behind, so nothing floats or sinks into the surface.
- **Color**, "keep the lighting": protects the overall palette and light direction, so it still looks like the same room.
- **Mask**, "paint only here": specifies exactly which pixels the model may change. Think of painter's tape.
- The first three maps are generated automatically. The mask needs one extra step.

6.2 Tuning the maps and building the mask

[Slides 11]



Fig. 5: Building the subtraction mask. Left to right: original frame; surface mask (countertop plus the materials on it); subtract mask (bucket, paper and hand); final mask (bare countertop only). Blue marks what Cosmos will repaint.

What we learned from tuning:

- **Edges matter most.** Too high and nothing changes; too low and objects warp.
- **Color guidance must be nearly off.** If the goal is to repaint something, this map fights the change.
- **Prompt strength controls boldness.** Higher settings give the model more confidence to commit.

How the mask is built (Fig. 5). The countertop and the bucket overlap; to change one without the other:

1. find the surface;
2. find what is on it;
3. subtract;
4. use the result as the final mask.

The bounding boxes come from the same scene model that built the ontology. No extra tools are needed.

7 Results

[Slides 12–16]

7.1 Simple aesthetic variations

[Slides 13]



(a) Old (b) New

Fig. 6: The models perform simple aesthetic variations fairly well: the countertop is recolored while the hand pose, bucket geometry and motion are retained.

– The models handle simple aesthetic variations fairly well (Fig. 6).
– To note in the text: the same output also shows the leakage described in Sec. 2 (wall, rag and bucket tone), which is the boundary explored next.

7.2 Multi-texture scenes

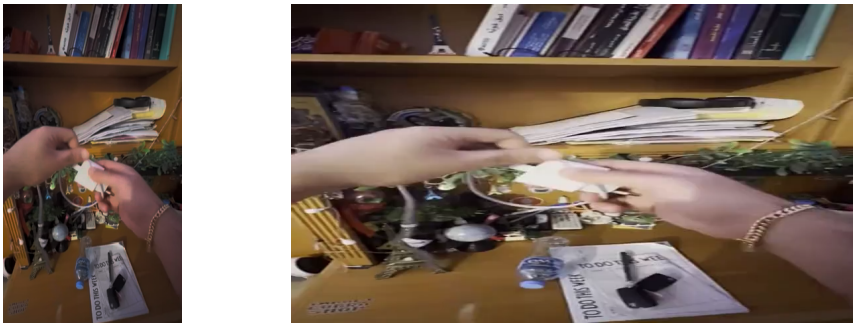
[Slides 14]

– The model struggles fairly heavily with a clip like Fig. 7, where the frame contains multiple textures.



(a) Old (b) New

Fig. 7: The model struggles heavily with a clip whose frame has multiple textures: nearly every surface is recolored, and the cushion patterns survive only as blue-on-blue relief.



(a) Original (b) Generated (Cosmos Predict)

Fig. 8: Physics-minded task: unravel a cord clockwise at constant angular velocity. The generated clip reproduces the scene without performing the motion.

7.3 A physics-minded task

[Slides 15]

– The model was asked to perform a physics-minded task: unravel a cord in a clockwise direction at constant angular velocity (Fig. 8).
– This is an action-level variation and was run with Cosmos Predict rather than Transfer.
– Outcome: the scene is reproduced; the cord does not move.

7.4 More results

[Slides 16]



(a) Shelf (b) Bed, rug (c) Paper rotation

Fig. 9: Further results. (a) Blue desk-and-shelf variant: color spreads onto shelf objects and the cable. (b) Bed-making clip with a rug texture prompt: surfaces smear and the hand warps. (c) Paper-rotation prompt with Cosmos Predict: no visible change.

– Three further renders (Fig. 9), each showing a distinct failure: color leakage beyond the target, geometric warping, and a no-op on an action prompt.

8 Closing the Gap and Future Directions

[Slides 17–18]

Two paths to close the gap:

– **Fine-tuning.** Teach Cosmos Predict the contact physics it lacks by fine-tuning on real manipulation video, with small focused adapters (LoRA) rather than retraining the whole model. If it works, it unlocks action-level synthetic data.
– **Cosmos 3.** The longer-term bet: Cosmos 3 collapses the three-model pipeline into a single architecture with a reasoning tower and a generation tower, with spatial reasoning built into the training data. The Nano 8B version fits on the A100 hardware already in use, but the tooling is very early [3,4].

What still needs to be done:

1. Scale to more than 100 clips with automated prompt generation.
2. Test whether appearance-only variations improve policy robustness. Everything so far is qualitative; this is the top priority.
3. Explore multi-view generation for a richer training signal.

9 Key Takeaways

[Slides 19]

1. A multi-model pipeline on NVIDIA Cosmos can turn a small real dataset into thousands of synthetic training clips.
2. Changing appearance works for simpler scenes. Complex scenes and action-level changes remain unsolved.
3. Structured reasoning about a scene produces better variations than random augmentation.

The open question: is synthetic appearance data enough to make a robot more robust, or do we need to solve action generation first?

References

1. NVIDIA: Cosmos-Predict2: A suite of diffusion-based world foundation models available in 2b and 14b. Hugging Face model card, https://huggingface.co/nvidia/Cosmos-Predict2-2B-Video2World (2025), code: https://github.com/nvidia-cosmos/cosmos-predict2. Accessed 2026-10-16
2. NVIDIA: Cosmos-Transfer2.5: A suite of diffusion-based world-to-world models. Hugging Face model card, https://huggingface.co/nvidia/Cosmos-Transfer2.5-2B (2025), released October 6, 2025. Code: https://github.com/nvidia-cosmos/cosmos-transfer2.5. Accessed 2026-10-16
3. NVIDIA: Cosmos 3: Omnimodal world models for physical AI. arXiv preprint arXiv:2606.02800 (2026), https://arxiv.org/abs/2606.02800, technical report also at https://research.nvidia.com/labs/cosmos-lab/cosmos3/technical-report.pdf
4. NVIDIA: Cosmos3-Nano. Hugging Face model card, https://huggingface.co/nvidia/Cosmos3-Nano (2026), released May 31, 2026, OpenMDW-1.1 license. Accessed 2026-10-16
5. NVIDIA, Abu Alhaija, H., Alvarez, J., et al.: Cosmos-Transfer1: Conditional world generation with adaptive multimodal control. arXiv preprint arXiv:2503.14492 (2025). https://doi.org/10.48550/arXiv.2503.14492, https://arxiv.org/abs/2503.14492
6. NVIDIA, Agarwal, N., Ali, A., et al.: Cosmos world foundation model platform for physical AI. arXiv preprint arXiv:2501.03575 (2025). https://doi.org/10.48550/arXiv.2501.03575, https://arxiv.org/abs/2501.03575
7. NVIDIA, Azzolini, A., Bai, J., et al.: Cosmos-Reason1: From physical common sense to embodied reasoning. arXiv preprint arXiv:2503.15558 (2025). https://doi.org/10.48550/arXiv.2503.15558, https://arxiv.org/abs/2503.15558

8. Tobin, J., Fong, R., Ray, A., Schneider, J., Zaremba, W., Abbeel, P.: Domain randomization for transferring deep neural networks from simulation to the real world. In: Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). pp. 23–30 (2017). https://doi.org/10.1109/IROS.2017.8202133